

```
Running on Kaggle
Kaggle: датасет знайдено у /kaggle/input/datasets/irustandi/yelp-review-polarity/yelp_review_polarity_csv.
Завантаження не потрібне.
Loading Yelp review data...
Successfully loaded 50001 entries.
Cleaning review content...
Text preprocessing complete.
Label distribution:
label
0  26832
1  23169
Name: count, dtype: int64
Splitting dataset into training and test sets...
Train size: 40000, Test size: 10001
Converting text to sequences...
Tokenization and padding completed.
    Vocabulary size : 10000
    X_train_pad shape: (40000, 100)
    X_test_pad shape : (10001, 100)
Constructing LSTM model...
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:97: UserWarning: Argument
`input_length` is deprecated. Just remove it.
    warnings.warn(
2026-05-22 23:01:47.761733: E external/local_xla/xla/stream_executor/cuda/cuda_platform.cc:51] failed
call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	?	0 (unbuilt)
lstm (LSTM)	?	0 (unbuilt)
dropout (Dropout)	?	0
lstm_1 (LSTM)	?	0 (unbuilt)

dropout_1 (Dropout)	?	0
dense (Dense)	?	0 (unbuilt)
dense_1 (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)

Trainable params: 0 (0.00 B)

Non-trainable params: 0 (0.00 B)

Training the model...

Epoch 1/5

625/625  **135s** 209ms/step - accuracy: 0.6220 - loss: 0.6243 - val_accuracy: 0.8917 - val_loss: 0.2727

Epoch 2/5

625/625  **130s** 207ms/step - accuracy: 0.9127 - loss: 0.2443 - val_accuracy: 0.9087 - val_loss: 0.2323

Epoch 3/5

625/625  **129s** 206ms/step - accuracy: 0.9461 - loss: 0.1602 - val_accuracy: 0.9020 - val_loss: 0.2577

Epoch 4/5

625/625  **129s** 207ms/step - accuracy: 0.9625 - loss: 0.1137 - val_accuracy: 0.8996 - val_loss: 0.2969

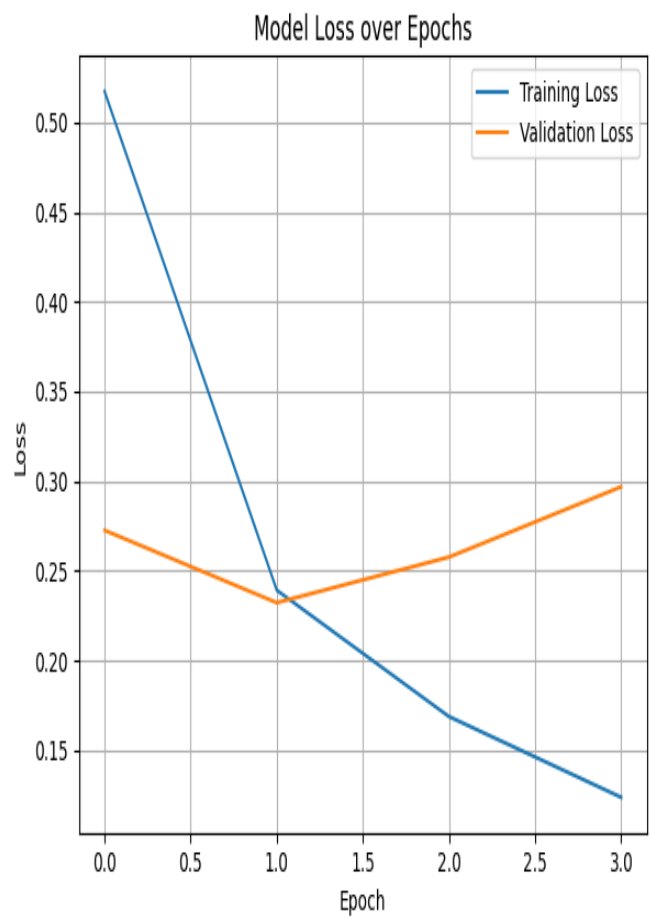
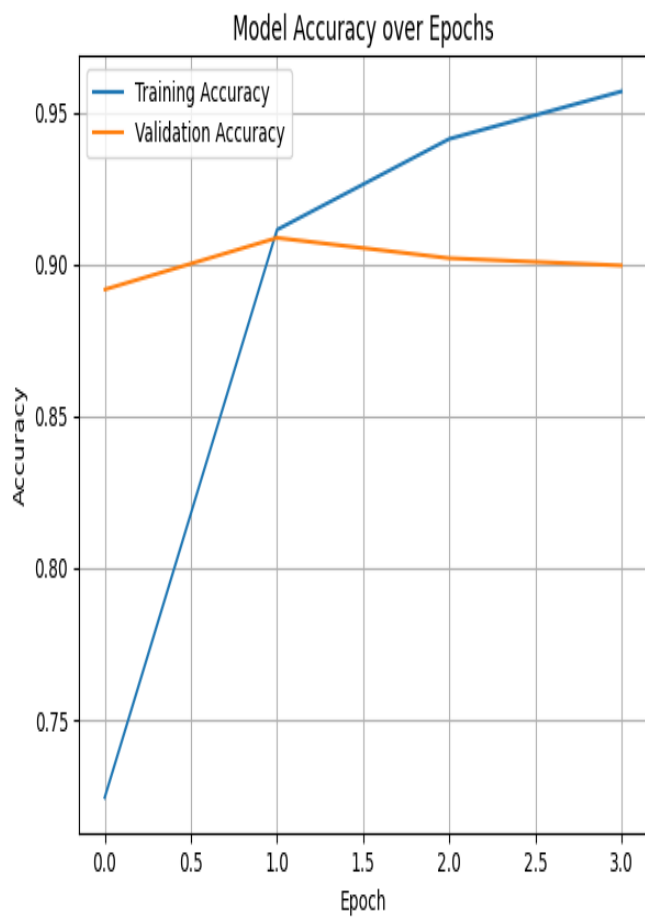
Epoch 4: early stopping

Restoring model weights from the end of the best epoch: 2.

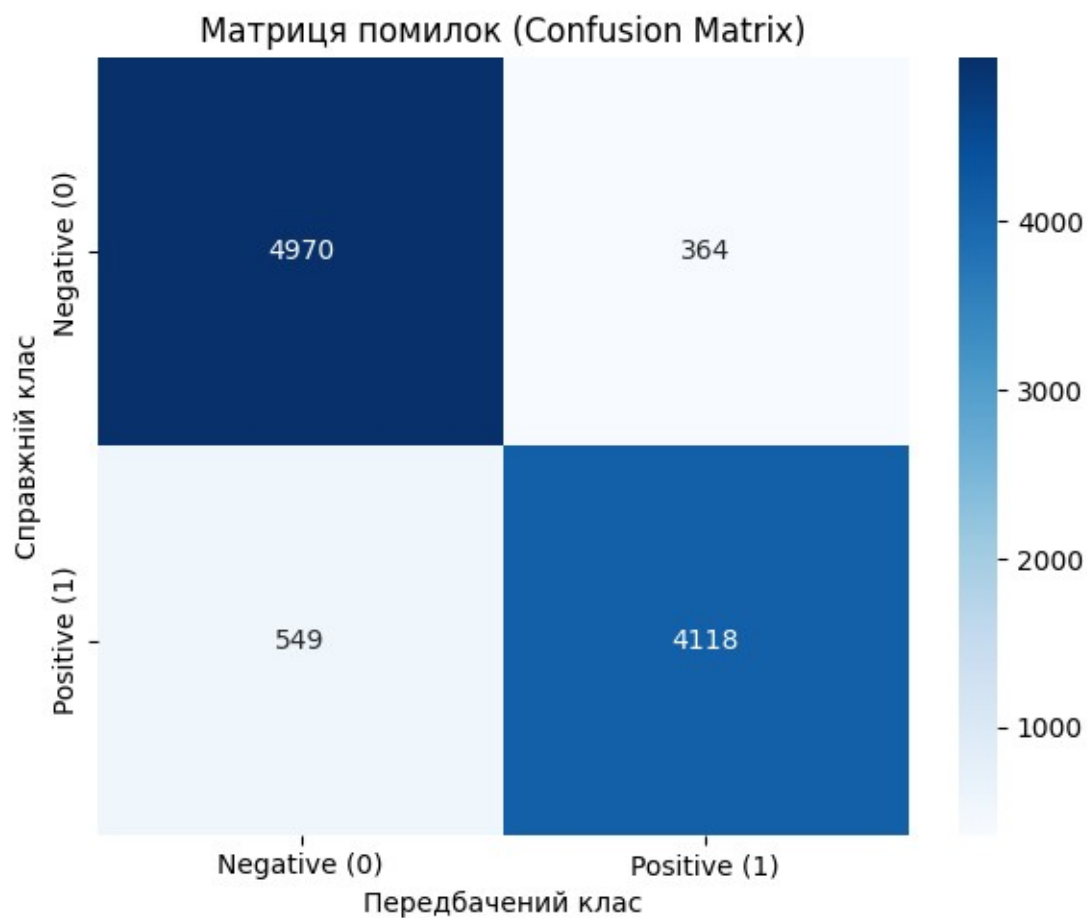
Evaluating on test data...

313/313  **16s** 52ms/step - accuracy: 0.9130 - loss: 0.2250

Test Set Accuracy: 0.9087



313/313 ————— **16s** 51ms/step



Accuracy : 0.9087

Precision : 0.9188

Recall : 0.8824

F-Score : 0.9002

Детальний звіт:

	precision	recall	f1-score	support
Negative (0)	0.90	0.93	0.92	5334
Positive (1)	0.92	0.88	0.90	4667
accuracy		0.91		10001
macro avg	0.91	0.91	0.91	10001
weighted avg	0.91	0.91	0.91	10001

Sample prediction tests:

Review: Absolutely loved the atmosphere and the food was fantastic!

Текст : Absolutely loved the atmosphere and the food was fantastic!...

Настрій : Positive (впевненість: 0.9781)

Sentiment: Positive

Review: I had to wait forever and the service was terrible.

Текст : I had to wait forever and the service was terrible....

Настрій : Negative (впевненість: 0.0167)

Sentiment: Negative

Review: The coffee was okay, nothing special but not bad either.

Текст : The coffee was okay, nothing special but not bad either....

Настрій : Negative (впевненість: 0.0148)

Sentiment: Negative

Review: Best restaurant I have ever visited! Will definitely come back!

Текст : Best restaurant I have ever visited! Will definitely come back!...

Настрій : Positive (впевненість: 0.9628)

Sentiment: Positive

--- Тести на прикладах з датасету ---

Текст : Unfortunately, the frustration of being Dr. Goldberg's patient is a repeat of th...

Настрій : Negative (впевненість: 0.0094)

[✓] Очікувано: Negative, Отримано: Negative

Текст : Been going to Dr. Goldberg for over 10 years. I think I was one of his 1st patie...

Настрій : Positive (впевненість: 0.9868)

[✓] Очікувано: Positive, Отримано: Positive

Текст : Wing sauce is like water. Pretty much a lot of butter and some hot sauce (franks...

Настрій : Positive (впевненість: 0.5287)

[X] Очікувано: Negative, Отримано: Positive

Текст : After waiting for almost 30 minutes to trade in an old phone part of the buy bac...

Настрій : Negative (впевненість: 0.0363)

[✓] Очікувано: Negative, Отримано: Negative

Текст : Don't waste your time. We had two different people come to our house to give us...

Настрій : Negative (впевненість: 0.0121)

[✓] Очікувано: Negative, Отримано: Negative

[1] Keras Tokenizer → Accuracy: 0.9087

[2] Токенізація за допомогою NLTK word_tokenize...

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Метод токенизації: NLTK word_tokenize

=====

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:97: UserWarning: Argument
`input\_length` is deprecated. Just remove it.

```
warnings.warn(  
Accuracy : 0.6507  
Precision : 0.5847  
Recall : 0.8684  
F1-Score : 0.6989
```

[3] Токенізація за допомогою SpaCy...

/usr/local/lib/python3.12/dist-packages/spacy/pipeline/lemmatizer.py:188: UserWarning: [W108] The rule-
based lemmatizer did not find POS annotation for one or more tokens. Check that your pipeline includes
components that assign token.pos, typically 'tagger'+attribute_ruler' or 'morphologizer'.

```
warnings.warn(Warnings.W108)
```

=====

Метод токенизації: SpaCy tokenizer

=====

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:97: UserWarning: Argument
`input\_length` is deprecated. Just remove it.

```
warnings.warn(  
Accuracy : 0.9028  
Precision : 0.9101  
Recall : 0.8785  
F1-Score : 0.8940
```

[4] BPE токенизація (HuggingFace tokenizers)...

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Метод токенизації: BPE (HuggingFace)

=====

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:97: UserWarning: Argument
`input\_length` is deprecated. Just remove it.

```
warnings.warn(  
Accuracy : 0.8897  
Precision : 0.8953  
Recall : 0.8648  
F1-Score : 0.8798
```

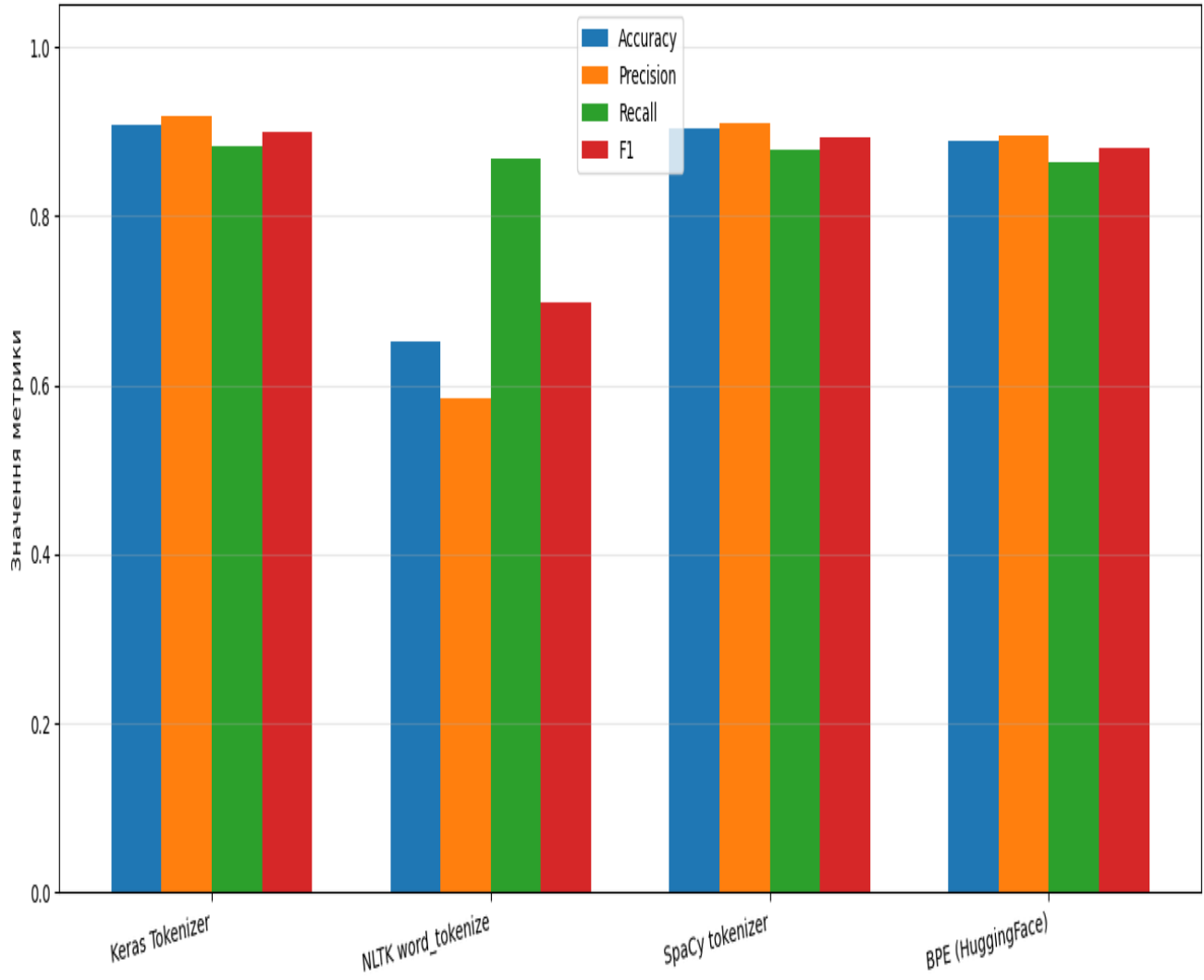
=====

ПОРІВНЯННЯ МЕТОДІВ ТОКЕНІЗАЦІЇ

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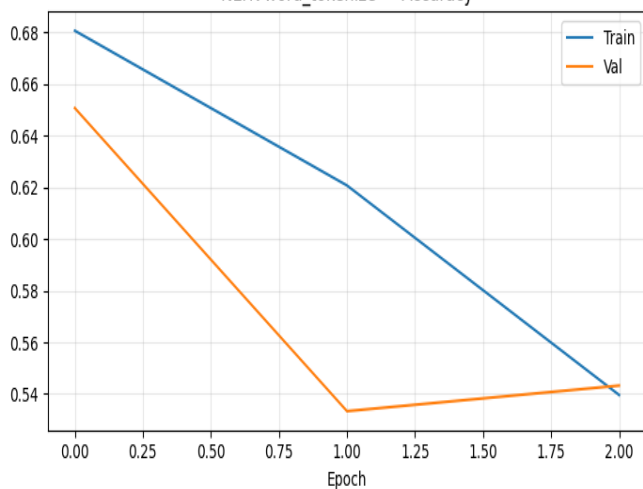
	method	accuracy	precision	recall	f1
	Keras Tokenizer	0.908709	0.918786	0.882366	0.900208
NLTK	word_tokenize	0.650735	0.584680	0.868438	0.698853
	SpaCy tokenizer	0.902810	0.910100	0.878509	0.894025
	BPE (HuggingFace)	0.889711	0.895297	0.864795	0.879782

Порівняння методів токенизації (LSTM, Yelp)

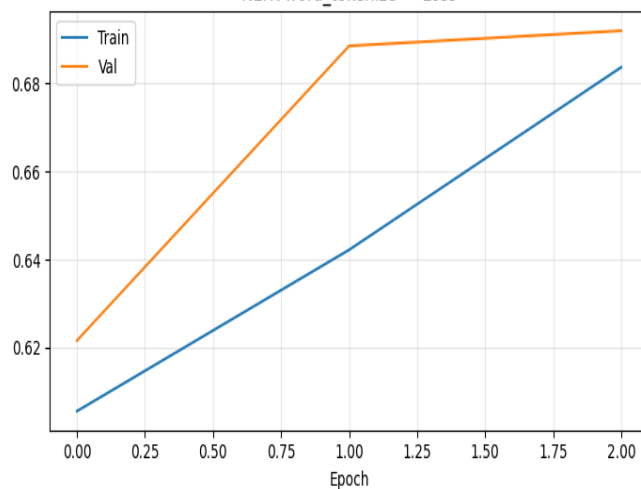


Криві навчання для різних методів токенизації

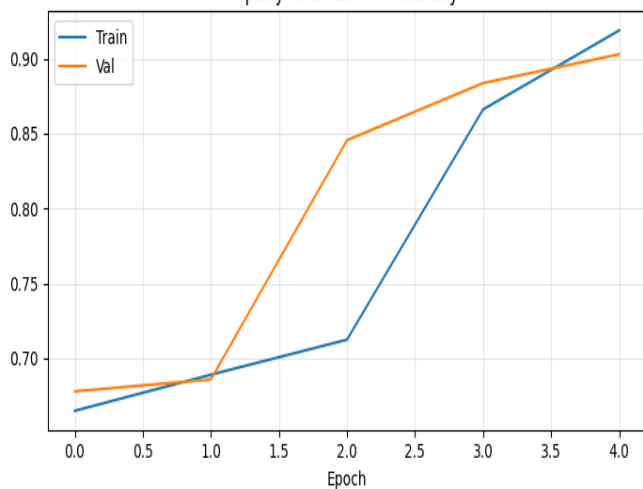
NLTK word_tokenize — Accuracy



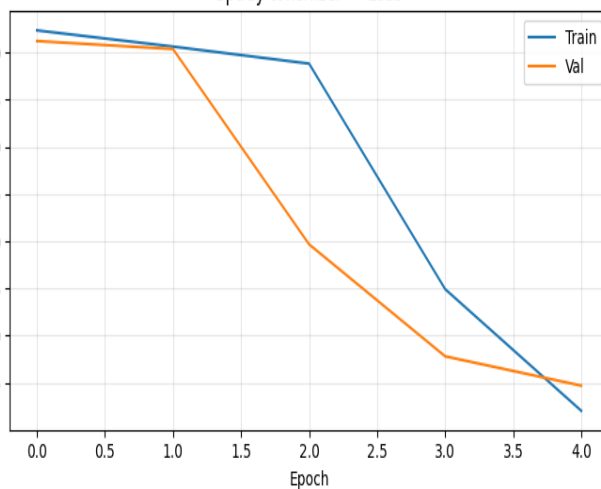
NLTK word_tokenize — Loss



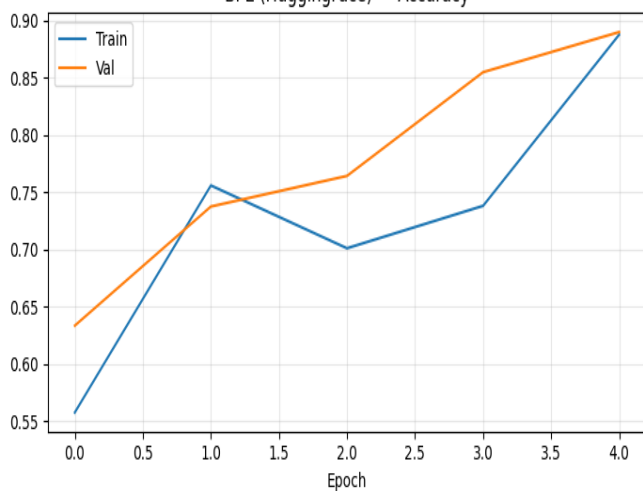
SpaCy tokenizer — Accuracy



SpaCy tokenizer — Loss



BPE (HuggingFace) — Accuracy



BPE (HuggingFace) — Loss

